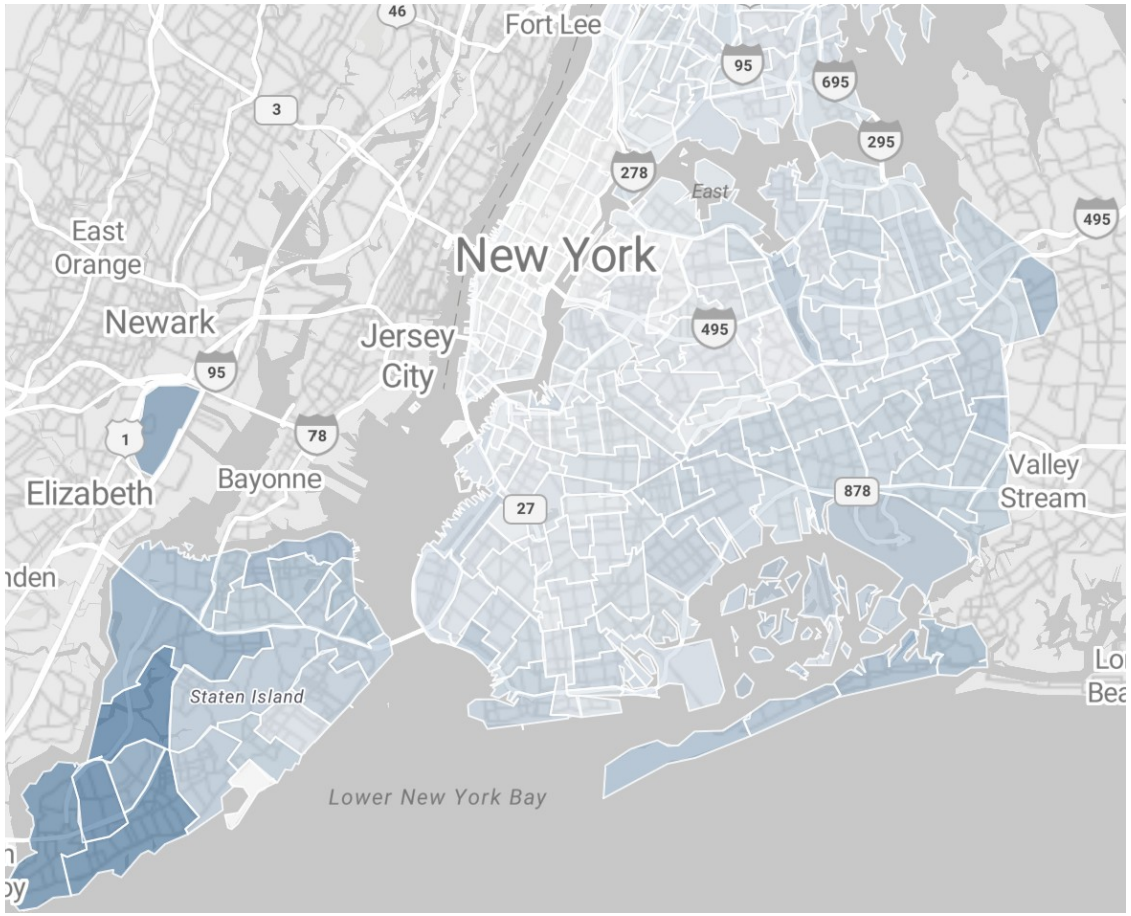
# NYC Yellow Taxi: when, where and how drivers earn

Power BI analysis of 898,131 trips

May–June 2020 · Angel Iana



# The data: 898,131 trips, checked in SQL and matched to TLC counts

**898,131 trips**
TLC trip records, 2 monthly files

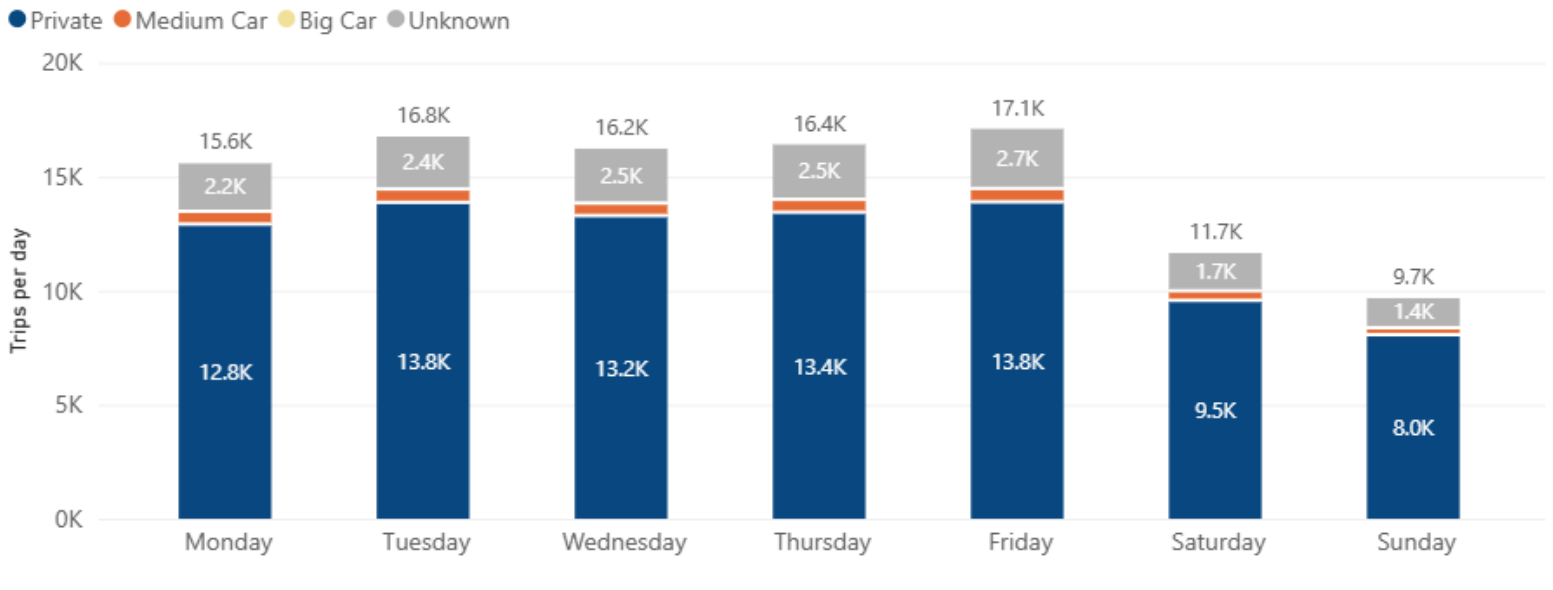
**SQL Server checks**
6,608 trips with impossible values flagged, not deleted

**Power BI report**
898,131 trips · all records

| | Our data | TLC official |
|---|---|---|
| Trips, May–June 2020 | 898,131 | 898,127 |
| Trips per day | 14,723.5 | 14,723.4 |

All records are kept, as required by the project brief. TLC figures: 11,237 trips per day in May and 18,326 in June. Every number in this deck comes from a SQL query and matches the Power BI report.

# Private cars are about 82% of trips every day; Friday and Tuesday are the busiest



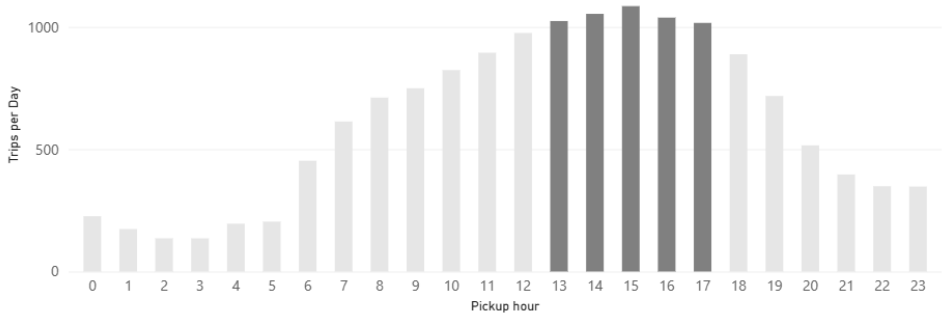

**80.8–82.7%**
Private, on every weekday

**17,100**
trips per day on Friday, the busiest

**9,672**
trips per day on Sunday, the quietest

The vehicle mix does not depend on the day; only the volume does.

# Demand peaks 13:00–17:00 on weekdays, with no morning rush




| weekday | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Monday | 210 | 143 | 117 | 116 | 198 | 238 | 523 | 721 | 838 | 882 | 914 | 974 | 1026 | 1084 | 1114 | 1148 | 1117 | 1119 | 926 | 692 | 479 | 378 | 319 | 309 |
| Tuesday | 195 | 148 | 127 | 138 | 210 | 242 | 548 | 738 | 895 | 972 | 1024 | 1101 | 1196 | 1189 | 1199 | 1227 | 1198 | 1152 | 978 | 744 | 509 | 379 | 331 | 323 |
| Wednesday | 190 | 149 | 125 | 148 | 220 | 238 | 533 | 727 | 878 | 906 | 987 | 1044 | 1092 | 1211 | 1183 | 1191 | 1110 | 1121 | 949 | 727 | 495 | 365 | 317 | 326 |
| Thursday | 197 | 164 | 133 | 155 | 222 | 240 | 562 | 737 | 867 | 893 | 983 | 1032 | 1141 | 1155 | 1173 | 1215 | 1153 | 1086 | 956 | 753 | 523 | 389 | 340 | 348 |
| Friday | 217 | 176 | 133 | 147 | 216 | 227 | 548 | 733 | 874 | 877 | 938 | 998 | 1106 | 1152 | 1191 | 1241 | 1225 | 1249 | 1084 | 872 | 607 | 454 | 405 | 430 |
| Saturday | 293 | 223 | 165 | 143 | 170 | 138 | 268 | 378 | 389 | 453 | 569 | 662 | 724 | 781 | 857 | 864 | 778 | 733 | 696 | 662 | 511 | 428 | 382 | 388 |
| Sunday | 274 | 207 | 150 | 108 | 138 | 113 | 210 | 286 | 273 | 296 | 386 | 483 | 574 | 633 | 686 | 738 | 705 | 674 | 644 | 580 | 484 | 381 | 343 | 306 |

**35.4%**

of trips start between 13:00 and 17:00

**1,085**

trips per day at 15:00, the peak hour

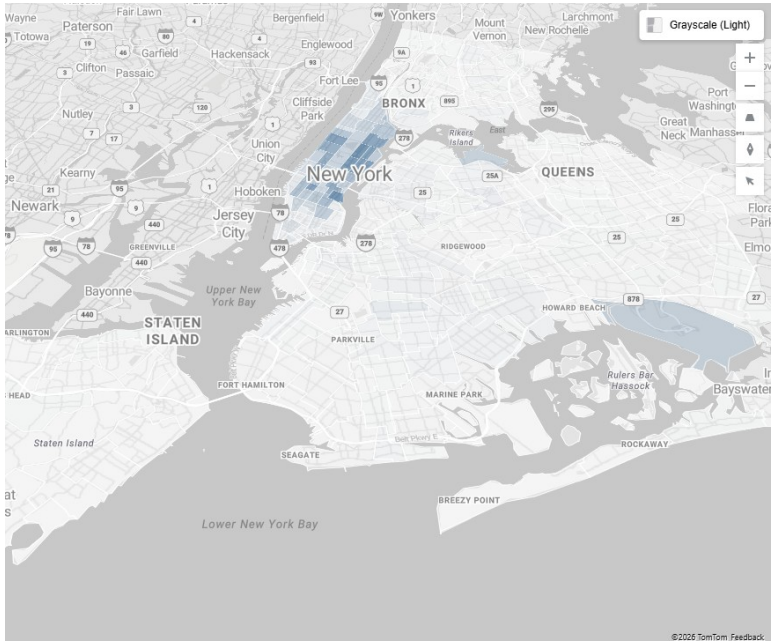
**1,249**

trips per day on Friday at 17:00, the busiest hour of the week

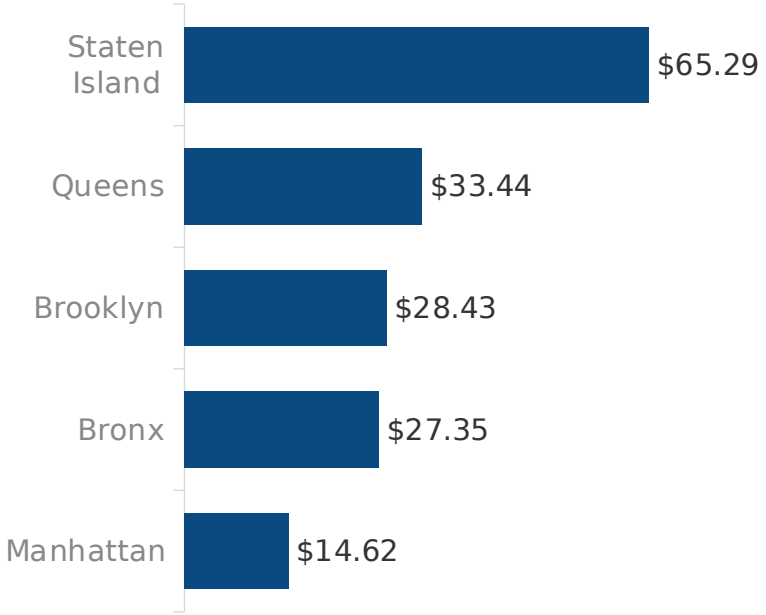
Most cars are needed on weekday afternoons, not at the usual commute times.

# Manhattan has 86% of pickups, but trips that start outside it pay about twice as much

**Pickups**




**Avg driver income per trip, by borough**



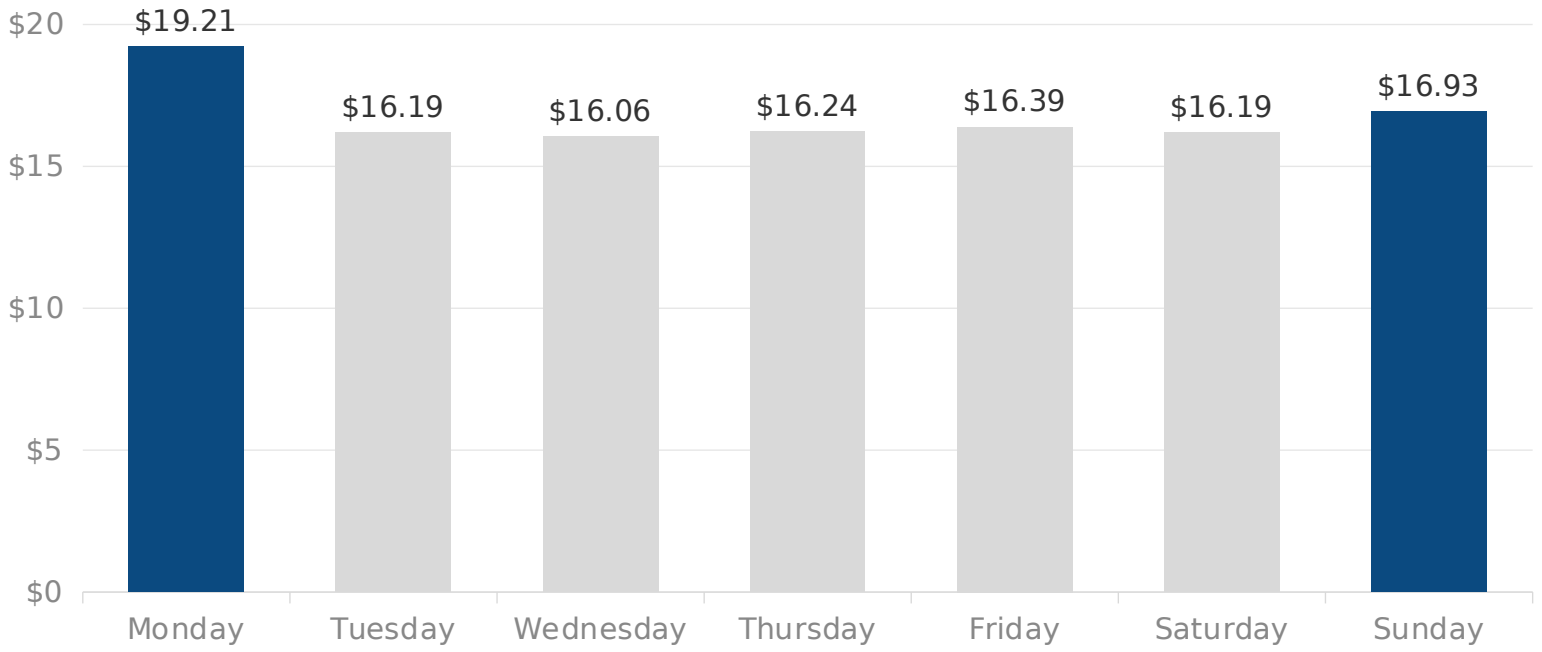

**86.33%**
of pickups are in Manhattan

**$33.44**
per trip from Queens, vs $14.62 from Manhattan

**$48.15**
per JFK pickup; airports are only 2.71% of pickups

Manhattan brings the volume; airports and outer boroughs bring the value per trip.

# Monday leads income per trip, but only because of one $429,562 trip




**$19.21**
average driver income per trip on Monday

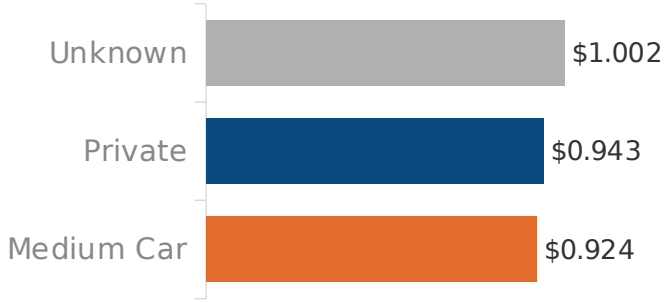
**$16.15**
on Monday without that single trip

**$16.93**
on Sunday, the second day

Without that one record, Sunday and Friday lead, and the gap between days is small.

# Tips per mile are about the same for every vehicle type once only card tips are counted

| Vehicle type | All records | Excl. extremes |
|---|---|---|
| Unknown | $1.002 | **$1.003** |
| Private | $0.943 | **$0.944** |
| Medium Car | $0.924 | **$0.925** |

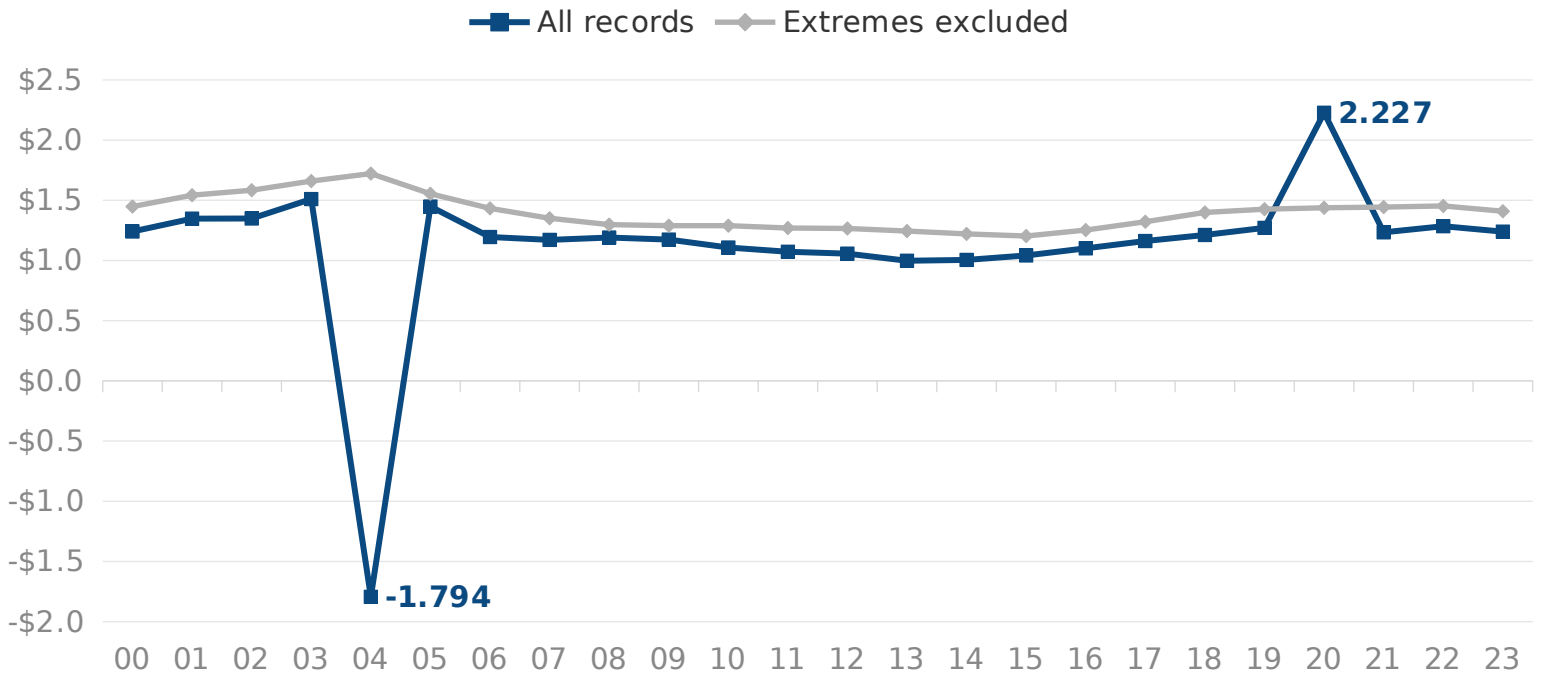



**$0.944**

tip per mile, all card payments

Cash tips are not recorded in TLC data, so only card payments are counted. Excluding the extreme values changes the result by less than 0.2%.

# The afternoon pays the least per minute; single records distort 04:00 and 20:00




**$1.061**

per minute 13:00–17:00, vs $1.358 at other hours

**−$1.794**

at 04:00: one record with a duration of −531,231 minutes

**$2.227**

at 20:00: the same $429,562 trip as on Monday

The busy afternoon hours are the slowest, so they pay the least per minute.

# What to do with these results

## Vendor

• Plan the most cars for weekday afternoons, 13:00–17:00 (35.4% of trips), and for Friday and Tuesday.
• Keep supply in Manhattan (86.33% of pickups); treat airports as few trips with high value ($48.15 per JFK pickup).
• Check trip data at the source: 6,608 records (0.74%) have impossible values, and one of them changes two answers.

## Driver

• Afternoon traffic pays the least per minute: $1.061 at 13:00–17:00 vs $1.358 at other hours.
• The JFK flat fare pays $58.85 per trip for 29.88 minutes (6,880 trips).
• Once the single $429,562 trip is set aside, the weekday matters little; the hour and the place matter more.

# Own initiative: the same analysis without the 6,608 extreme records

## Why

• A few impossible values can move averages a lot, so I also looked at the answers without them.
• The records are flagged in a Data Quality Flag column, not deleted, and filtered on separate report pages marked (excl.).

## Answers without the extreme values

• Best income days: Sunday ($17.07) and Friday ($16.49).
• Best hour per minute: 04:00 ($1.722); worst: 15:00 ($1.203).
• Busiest days, peak hours, tips and zones: the same as with all records.
• Full deck for this view: presentation/B_extremes-excluded.

| Flag | Rule | Records |
|---|---|---|
| negative_fare | fare below $0 | 4,062 |
| duration_gt_300 | trip longer than 5 hours | 1,281 |
| duration_le_0 | duration of 0 minutes or less | 1,211 |
| distance_gt_200 | distance above 200 miles | 30 |
| out_of_period | date outside May–June 2020 | 23 |
| fare_gt_1000 | fare above $1,000 | 1 |
| **Total** | **0.74% of 898,131 records** | **6,608** |

# Comparison: 4 of 6 answers stay the same; 2 change because of single records

| Question | All records | Extremes excluded | Changed |
|---|---|---|---|
| 1 Busiest days | Friday 17,099.7 · Tuesday 16,761.6 | Friday 16,978 · Tuesday 16,643 | No |
| 2 Peak hours | 13:00–17:00 (35.44%) | 13:00–17:00 (35.44%) | No |
| 3 Best income days | Monday $19.21 · Sunday $16.93 | Sunday $17.07 · Friday $16.49 | Yes – one $429,562 trip |
| 4 Tip per mile (card) | Unknown $1.002 · Private $0.943 | Unknown $1.003 · Private $0.944 | No |
| 5 Zones | Manhattan 86.33% · Kips Bay 35,327 | Manhattan 86.37% · Kips Bay 35,081 | No |
| 6 Income per minute | Best 20:00 $2.227 · 04:00 negative | Best 04:00 $1.722 | Yes – one −531,231-min trip |

- Totals change little: +0.7% trips per day, +2.9% farebox per day.
- Averages over time and distance move most: minutes per trip +10.1%, average speed +49.4%, income per minute −7.1%.
- Δ% = (all records − extremes excluded) / extremes excluded. All values come from SQL and match the Power BI comparison page.

# Limits of this analysis and next steps

## Limits

• The flag rules catch impossible values, not every suspicious one: a $950 trip of 0.3 miles passes every rule.
• Cash tips are not recorded, so tip results use card payments only.
• Minutes are trip time only; waiting time between trips is not in the data.
• There is no vehicle or driver ID, so income per shift or per car is unknown.
• May–June 2020 was the start of the recovery after the lockdown; the results describe that period only.

## Next steps

• Repeat the analysis on a full year of TLC data to compare with normal demand.
• Add plausibility rules, such as fare per mile, to catch suspicious trips and not only impossible ones.